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L.P.H Nethmi

IT23580862

Deep Learning-Based Intrusion Detection and Threat Mitigation in Healthcare Systems

Author : Liyana Patabendige Himara Nethmi (BSc in Cyber Security(UG))

Abstract - The accelerated digitalization of healthcare systems has significantly improved patient care services through electronic health records, cloud computing, and Internet of Medical Things (IoMT) devices. However, the increased connectivity has expanded the attack surface, making healthcare environments vulnerable to cyber-attacks such as distributed denial-of-service (DDoS) attacks, ransomware, malware, and advanced persistent threats. Conventional intrusion detection systems employ signature-based or rule-based approaches, which are inefficient against complex and zero-day attacks. Deep learning techniques have been recognized as effective tools owing to their capability to automatically learn patterns from large-scale network traffic data. This paper presents a literature review of deep learning-based techniques for intrusion detection and mitigation in healthcare systems, analyzing models such as convolutional neural networks (CNN), recurrent neural networks (RNN), long short-term memory (LSTM), autoencoders, and deep reinforcement learning. The review analyzes the detection accuracy, datasets, computational complexity, and practical implementation difficulties in real-world healthcare settings. In addition, it highlights important research gaps, including the lack of healthcare-specific datasets, high computational complexity, lack of interpretability, and difficulties in real-time mitigation. The results show that although deep learning techniques greatly enhance intrusion detection capabilities, more efficient, scalable, and interpretable models are required for practical deployment.

Index Terms— Intrusion Detection System (IDS), Deep Learning, Internet of Medical Things (IoMT), Healthcare Cybersecurity, Threat Detection, Threat Mitigation, Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Reinforcement Learning, Network Security.

I. Introduction

The growing pace of technological advancements has led to a substantial shift in the current healthcare infrastructure. The convergence of electronic health records (EHRs), cloud computing infrastructure, artificial intelligence, and the Internet of Medical Things (IoMT) has made it possible to monitor patients, diagnose them, and make clinical decisions in real-time. IoMT devices like wearable sensors, implantable devices, and intelligent monitoring systems are constantly collecting and transmitting confidential patient information over a network of interconnected devices [2]. Although these technologies improve the efficiency of the healthcare system and patient outcomes, they also pose severe cybersecurity threats.

The current healthcare infrastructure has become a lucrative target for cybercriminals due to the immense value of medical information and the importance of healthcare services. Cyber-attacks like distributed denial-of-service (DDoS) attacks, ransomware attacks, malware injection attacks, data breaches, and advanced persistent threats can cause disruptions in healthcare services, compromise patient confidentiality, and even threaten human lives [7]. The growing connectivity of the healthcare infrastructure has increased the attack surface, making it impossible to secure the infrastructure using traditional security solutions.

Intrusion Detection Systems (IDS) are critical for the detection of malicious activities in the healthcare network. Traditional IDS solutions are usually based on signature-based or rule-based systems. Even though these systems are effective in the detection of known threats, they are not very effective in the detection of zero-day attacks and more complex evolving threats. In addition, the dynamic and heterogeneous nature of IoMT environments makes it even more difficult to provide accurate threat detection.

In the last few years, deep learning solutions have appeared as a promising alternative for cybersecurity challenges [5]. Solutions such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), autoencoders, and reinforcement learning-based systems have shown a superior ability to learn complex traffic patterns and detect anomalies in large datasets.

Despite the progress made, there are still some challenges that need to be overcome in the application of deep learning-based IDS in real-world healthcare environments. These include high computational complexity, lack of interpretability, limited availability of healthcare-oriented datasets, and the inability to provide real-time protection against threats. Overcoming these challenges is crucial for the development of secure, scalable, and reliable healthcare-oriented cybersecurity solutions.

It is against this background that this paper seeks to provide a comprehensive literature review of deep learning-based intrusion detection and threat mitigation strategies in healthcare environments.

II. Research Objectives

This study examines the existing literature on the application of deep learning techniques in the healthcare system for the detection and mitigation of network attacks. This paper critically analyzes existing research contributions, their effectiveness, and identifies the gaps in the field of cybersecurity for healthcare systems.

The major aim of conducting this paper is to assess the role of deep learning-based intrusion detection systems in the mitigation of cyber attacks in Internet of Medical Things (IoMT) environments.

To achieve this objective, the study addresses the following research goals:

1. To analyze different deep learning architectures such as Convolutional Neural Networks, Recurrent Neural Networks, Long Short-Term Memory, Autoencoders, and hybrid architectures that integrate CNN and LSTM, and reinforcement learning that can be used for developing healthcare intrusion detection systems.

2. To evaluate different benchmark datasets used for developing healthcare cybersecurity, their suitability, and their effectiveness for real-world environments.

3. To compare different intrusion detection systems developed based on deep learning architectures and compare their performance metrics such as accuracy, precision, and false positive rate, and computational efficiency.

4. To analyze different integrated threat mitigating strategies such as adaptive learning, fog and edge computing, federated learning, and privacy-preserving security.

5. To identify different challenges and research gaps that prevent the development and implementation of deep learning-based security frameworks for healthcare infrastructures.

6. Overall, this research study will provide an understanding of different research and developments that can be used for developing real-time intrusion detection systems for healthcare environments.

III. Literature Review

A. Cybersecurity Threats in Healthcare Systems

The healthcare industry has turned out to be one of the primary industries that are frequently targeted by cybercriminals due to the nature of medical data and the significance of the services provided. Today, the healthcare industry is highly dependent on digital technologies such as Electronic Health Records, cloud computing, and IoMT devices. However, it also increases the risk of cybercrime.

The emergence of the Internet of Medical Things has increased the risk of cybercrime due to the communication of medical devices, hospital networks, and cloud environments. It enables the exchange of sensitive patient data among medical devices, hospital networks, and cloud environments. This increases the risk of cybercrime in the healthcare infrastructure.

The cybercrime that frequently hits the healthcare infrastructure is ransomware, data breaches, hijacking of medical devices, and network-based cybercrime such as Distributed Denial of Service. Ransomware causes disruptions in the hospital environment due to the restriction of access to patient data. Data breaches compromise the privacy of patients, which leads to legal consequences. Hijacking medical devices enables the attacker to access medical devices, which compromises the safety of patients. Distributed Denial of Service causes disruptions in the hospital network.

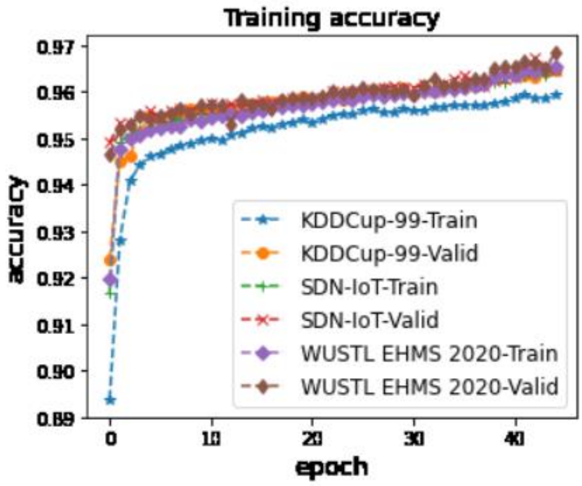

Fig 1: Common Cyber Attacks in Healthcare Networks

B. Traditional Intrusion Detection Systems (IDS)

Traditionally, intrusion detection systems have been used to effectively monitor the traffic in the network and detect any suspicious activities. Intrusion detection systems use rule-based IDS to detect abnormal activities in the network. These systems typically rely on:

1. **Signature-based detection**, where known attack patterns are matched against network traffic.
2. **Rule-based detection**, where predefined rules identify abnormal behavior.
3. **Firewalls**, which act as the first line of defense by blocking unauthorized access.

Although effective against known threats, these approaches have limitations. They cannot detect unknown or zero-day attacks and struggle to adapt to the dynamic nature of healthcare IoMT environments.

Additionally, firewalls can be used as the first line of defense in the healthcare network.

However, there are many disadvantages associated with the traditional IDS systems. Traditional IDS systems can only protect the healthcare networks from the known threats. They cannot protect the networks from the new threats that might arise in the future. Furthermore, traditional IDS systems cannot adapt to the changing environment of the IoMT systems. The traditional security systems cannot be used in the advanced healthcare systems.

C. Machine Learning-Based IDS

Machine learning-based intrusion detection systems were developed to mitigate the shortcomings associated with the traditional approach. This system relies on the ability of the machine to learn from the normal and anomalous behavior observed from the network data.
Machine learning algorithms help in the analysis of behavior and detection of anomalies. This enables the system to recognize unknown intrusion attempts. This approach False positives represent another challenge increases the accuracy of the detection process.

The machine learning model also requires the process of feature engineering. Experts need to choose the appropriate features from the network data before the model can be trained. This makes the process more complex. It also makes the system less flexible.

D. Deep Learning-Based Intrusion Detection

Deep learning techniques provide significant improvements over traditional and machine learning-based IDS by automatically learning complex patterns from large datasets.

1. **Deep Neural Networks (DNN)**

   Deep Neural Networks have the capability to detect abnormal network traffic patterns by learning hierarchical representations. Deep Neural Networks can be used to increase the accuracy of classification.

2. **CNN-Based IDS**

   Convolutional Neural Networks (CNN) have been employed for Intrusion Detection Systems due to their capability to learn meaningful features from network traffic data [1]. CNN-Based IDS can classify different types of attacks with better detection accuracy.

3. **RNN and LSTM-Based IDS**

   Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) have been employed for Intrusion Detection Systems due to their capability to analyze sequential data with temporal dependencies [10]. RNN-Based IDS can detect time-dependent attacks like DDoS and botnet attacks.

   Hybrid models using CNN and LSTM have been proposed to increase the accuracy of detection.

Compared to traditional intrusion detection approaches, deep learning-based models provide improved accuracy and anomaly detection capabilities. However, they often require higher computational resources and large training datasets, limiting real-time deployment in IoMT environments.

E. Deep Learning Architectures Used in Healthcare IDS

Different types of neural network architectures are utilized for intrusion detection systems based on deep learning techniques to detect malicious activity in healthcare network traffic.

Deep Neural Network (DNN)
DNN models contain several hidden layers that enable the system to learn hierarchical representations of data. This type of model helps to improve the accuracy of intrusion detection systems.

Convolutional Neural Network (CNN)
CNN models are effective in learning significant features from high-dimensional data, such as network traffic flows. In a healthcare environment, a CNN-based IDS system is able to classify types of cyber attacks, including malware, ransomware, and DDoS attacks.

Another type of model that has been employed is Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) models. These models have been used to analyze the sequential data and identify patterns in the behavior of the network. This feature makes the models appropriate for detecting time-dependent attacks such as botnet and slow network intrusions.

Hybrid models that use the combination of CNN and LSTM have been proposed. These models leverage the feature extraction capability of the CNN and the sequence learning capability of LSTM.

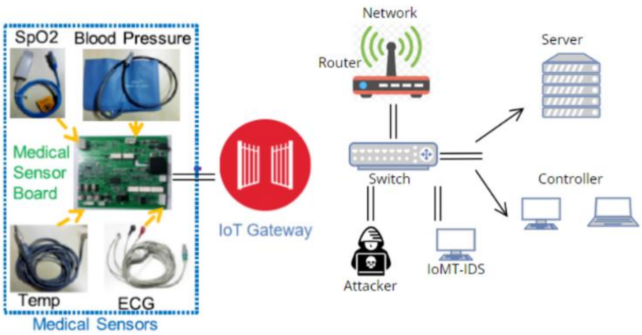

Fig 2: Deep Learning-Based Intrusion Detection Architecture

F. Deep Learning in Healthcare Networks

Deep learning is being used extensively in healthcare networks to safeguard sensitive patient information and provide secure communication between medical devices.

Healthcare environments produce massive amounts of data from IoMT devices, cloud computing, and hospital networks. Deep learning algorithms process this information to detect irregular activity and potential threats.
Deep learning models can analyze network traffic
The main use cases are:

- Protecting patient records
- Securing communication between medical devices
- Detecting compromised IoMT devices
- Monitoring hospital network activity

These capabilities improve the overall security of healthcare infrastructures.

G. Role of IoMT in Expanding Attack Surface

The Internet of Medical Things is an integral part of modern health care infrastructures because of its real-time monitoring and diagnostic capabilities. However, the increased interconnectedness of devices also poses a greater threat.

IoMT devices such as wearable devices, infusion pumps, and implantable devices constantly communicate sensitive information through the network.

These devices are usually not powerful computers and may not have adequate security features. Therefore, they become a point of entry for hackers.

The interconnectedness of IoMT devices makes it easier for hackers to move through the network once access is gained. Therefore, the following risks are presented:

- Data leakage
- Service disruption
- Manipulation of devices

The use of deep learning-based IDS can help in the monitoring of the communication between the IoMT devices.

H. Threat Detection Using Deep Learning

Threat detection is defined as the identification of malicious activities within a network before they result in damage. This is important in a healthcare environment due to the potential disruption that can be caused by cyber attacks.

Healthcare organizations have a considerable amount of network traffic due to the presence of multiple medical devices, cloud services, and digital patient records. However, conventional security systems cannot detect advanced attacks such as zero-day attacks and insider attacks. Therefore, deep learning-based intrusion detection systems have been developed to counter this problem.

Deep learning models can analyze network traffic patterns and detect abnormal behavior in healthcare IoMT environments [4], [7], [13]. These systems enable the detection of different cyber attacks, such as:

- Distributed Denial-of-Service (DDoS) attacks
- Malware infections
- Botnet activities
- Insider threats
- Data injection attacks

Deep learning models such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) can be used to learn complex patterns using vast amounts of data. CNN models can be used for classifying different kinds of attacks, while LSTM models can be used to learn patterns based on time.

These systems monitor network traffic patterns to detect attacks more efficiently.

I. Real-Time Intrusion Detection in Healthcare

Healthcare infrastructures must respond quickly to cyber attacks.

Deep learning-based IDS provide real-time monitoring through continuous traffic analysis.

Consequences of not having real-time detection capabilities are:

- Data breaches
- Compromise of devices
- Service disruptions

This is especially true in environments such as intensive care units.

J. Threat Mitigation Using Deep Learning

Whereas threat detection is concerned with detecting threats, threat mitigation is concerned with taking actions that reduce or prevent the effects of detected threats.

For a healthcare system, mitigation is an essential aspect of handling cyber threats, considering that delayed actions may have a negative impact on patient care and services. The use of deep learning in a system is essential in developing automated mitigation strategies that include developing real-time response mechanisms.

The use of reinforcement learning in a system enables it to respond dynamically to cyber threats [8]. The system is able to learn from network activities and respond appropriately by choosing an effective response from a number of options, such as:

- Isolating compromised devices
- Blocking malicious traffic
- Updating security policies
- Redirecting suspicious data flows

For example, in a situation where a DDoS attack is detected, mitigation mechanisms are able to respond in real time by limiting traffic from suspicious sources. Another example is that if a compromised device is detected in a healthcare system, mitigation mechanisms are able to respond by isolating the device from the network.

The use of automated mitigation in a system makes it more resilient and able to respond more efficiently in situations of cyber attacks.

Advanced mitigation systems can automatically respond to detected threats by isolating compromised devices, blocking malicious traffic, and updating security policies. Reinforcement learning-based approaches enable adaptive response strategies that improve system resilience in real-world healthcare deployments.

Advanced threat mitigation capabilities go beyond the simple detection of threats by also providing the capability for automated response actions. This is because deep learning-based systems can automatically isolate compromised systems, stop malicious traffic flows, and update firewall rules.

Reinforcement learning-based decision systems can also be used for more effective threat mitigation by choosing the best course of action for response based on observed network behavior. This is because such systems learn from past attack patterns.

In the case of real-world healthcare environments, such smart threat mitigation systems are critical for ensuring that clinical services are not disrupted while patient data is also protected against cyber threats.

K. Integration of Deep Learning with Edge and Fog Computing

In healthcare networks, there may be a need for the implementation of real-time threat detection. However, there may be latency in the centralized cloud environment while processing large amounts of data.

Edge and fog computing provide the following benefits [2]:

Analysis of data at the source using distributed processing.

Integrating deep learning with edge computing offers the following benefits:

- Faster threat detection
- Reduced latency
- Real-time response

Fog computing offers the following benefits:

- Collaborative threat detection using multiple devices in healthcare networks.

L. Challenges in Deep Learning-Based IDS for Healthcare

Despite the major breakthroughs realized in the development of deep learning-based intrusion detection systems, there are still some challenges in the implementation of these systems in the healthcare sector.

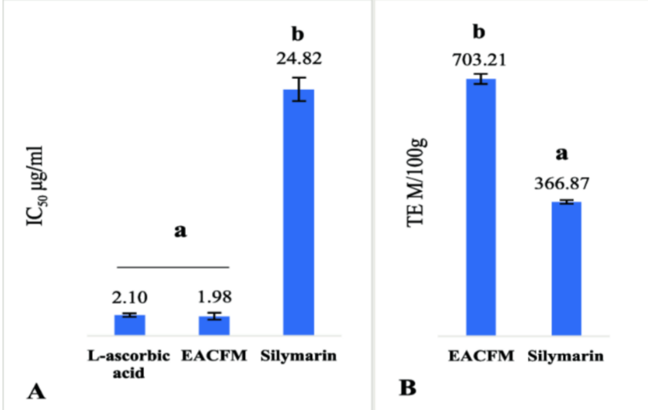

Fig 3: Security Challenges in IoMT Environments

Figure 3 highlights the major challenges faced when deploying deep learning-based intrusion detection systems in healthcare environments.

One of the major challenges is the high computational complexity of deep learning models. Healthcare networks, especially those that involve IoMT devices, are resource-constrained. Most healthcare devices have limited processing capabilities and memory. This makes it difficult to implement complex deep learning models on these devices.

Another major challenge is the need for large amounts of training data. Deep learning models require a huge amount of data to be able to perform with high accuracy. However, there is a lack of healthcare-related cybersecurity data due to privacy and regulatory issues.

Privacy is also a major concern [6]. Healthcare organizations deal with sensitive patient data. Collecting network traffic data for training models can lead to the leakage of sensitive patient data. This makes it difficult to ensure data security during the training of detection models.

False positives represent another challenge. Deep learning-based IDS can sometimes misinterpret normal traffic as malicious. This can lead to unnecessary alerts. In the healthcare sector, high false positives can lead to inefficiencies in critical services.

Real-time deployment is also challenging. Healthcare systems require immediate response to cyber threats, but deep learning models may introduce latency during detection and decision-making processes.

M. Scalability of Deep Learning-Based IDS

Healthcare networks vary in size from small clinics to large hospital infrastructures.
Deep learning-based IDS must be scalable to operate efficiently across different environments.
Scalable models ensure:

- Efficient performance
- Reduced computational load
- Compatibility with IoMT devices

N. Limitations of Existing Research

Despite the fact that many researchers have worked to find the best solution for the problem of intrusion detection in health care systems through the application of deep learning techniques, some weaknesses have been identified in the current research field.

First off, the main focus in most research is the detection of intrusions in health care systems rather than the mitigation of the threats. Although the detection of the threat is important, the mitigation or the response to the threat is also important in health care systems.

Secondly, some weaknesses have also been identified in the datasets that are used to evaluate the models in the field. Most models have been tested with generic datasets that may not represent the real-world scenario in health care systems.

The next weakness that has been identified in the field is the fact that the models are not implemented in real-time systems. Most models have been tested in simulated environments. Thus, the real-world scenario is not well understood.

Moreover, the models that have been developed in the field are not resource-friendly. Thus, the models may not be easily implemented in resource-constrained IoMT environments.

Finally, the evaluation metrics used in the field have not been standardized.

O. Types of Cyber Attacks in Healthcare Networks

The healthcare sector faces various types of cyber threats because of its interconnected systems and IoMT devices.

The most common cyber threat in this sector is ransomware attacks. These attacks involve the encryption of hospital data and demand a payment in return for decryption. These attacks can interfere with medical operations and slow down the treatment process.

The second major threat in this sector is data breaches. These breaches occur when an unauthorized person accesses confidential patient information. This can lead to financial and reputational losses.

Distributed Denial-of-Service attacks overwhelm the healthcare network with traffic. This causes a lack of access to healthcare services.

In addition, the sector faces hijacking of devices. This involves gaining control of medical devices such as infusion pumps and pacemakers.

These attacks emphasize the need for intelligent intrusion detection systems.

P. Deep Learning-Based Detection Workflow

In general, intrusion detection systems based on deep learning have a particular workflow to identify cyber attacks.

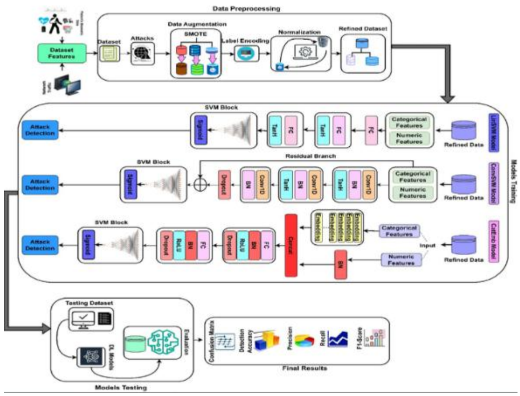

Fig 4: Deep Learning-Based Intrusion Detection Workflow in IoMT Environments

Figure 4 illustrates the workflow used by deep learning models to detect cyber threats in healthcare networks.

To start with, data is gathered from health networks, IoMT, servers, and cloud environments.

After that, data is preprocessed to eliminate any unwanted information.

Then, data is fed to a model for training purposes.

In this context, deep learning models such as CNN and LSTM learn how to recognize normal or malicious behavior.

Later on, real-time network traffic is examined to detect any anomalies that could be a cyber attack.

Overall, this particular workflow helps to detect cyber attacks in a timely manner.

Q. Deep Learning-Based Mitigation Workflow

Once a threat is detected, mitigation mechanisms are activated to minimize impact.
Mitigation involves:

- Identifying compromised devices
- Isolating malicious traffic
- Updating firewall rules
- Blocking unauthorized access

Reinforcement learning models can dynamically determine the best response based on network behavior.
This automated response improves healthcare network resilience.

R. Comparison of Traditional IDS and Deep Learning-Based IDS

Feature	Traditional IDS	Deep Learning IDS
Detection Capability	Known attacks	Known + Unknown
Feature Extraction	Manual	Automatic
Adaptability	Low	High
Detection Accuracy	Moderate	High
Real-Time Monitoring	Limited	Effective

Deep learning-based IDS provide improved accuracy and adaptability.

IV. Future Research Directions

Considering the rapid changes in healthcare technologies and the growing usage of IoMT devices in the healthcare sector, it is essential to develop advanced cybersecurity systems. Although the use of deep learning algorithms in intrusion detection systems has been found to be effective in the past, there are several areas that need to be considered in the future to improve the efficiency of the proposed systems in the real world.

A new direction for future research is the development of efficient deep learning algorithms. Most of the proposed algorithms in the past are computationally intensive, making it difficult to implement them in the real world. Therefore, it is essential to develop efficient algorithms that can be implemented in the real world with high efficiency in terms of detection accuracy while minimizing the complexity of the algorithms used in the proposed systems.

Another critical direction for future research is the development of specific datasets for the proposed systems. In most of the proposed systems in the past, the datasets used were related to the IoT environment, which is not specific to the healthcare environment.

Future research should also be conducted to investigate real-time mitigation of threats. While many research papers have been written about detecting threats, fewer papers discuss how to respond to threats. The integration of deep learning with other automatic response techniques may improve the robustness of healthcare systems.

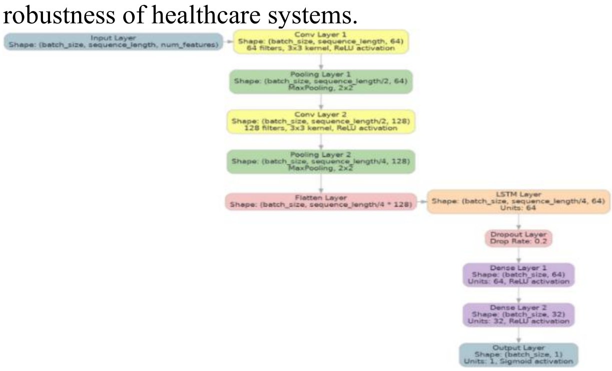

Fig 5: Hybrid CNN–LSTM Architecture for Real-Time Intrusion Detection

Furthermore, the integration of reinforcement learning may improve adaptive security solutions. This will allow the solutions to learn and adapt to changing cyber threats and minimize human intervention.

Another area of research for improving cybersecurity solutions for the healthcare domain is the integration of privacy-preserving techniques. The security of patient information during the training of detection models remains a challenge. The integration of federated learning may improve the security of patient information [13].

Finally, the focus in the future should be on explainable deep learning models. Understanding the decision-making process is critical in developing trust in the system for healthcare applications.
Advancements in these areas are important in the development of scalable, efficient, and secure intrusion detection systems for the healthcare industry.

This review highlights the importance of integrating real-time mitigation with deep learning-based detection in healthcare networks, an area that remains underexplored in current intrusion detection research.

Future research should focus on lightweight deep learning models suitable for resource-constrained IoMT devices. In addition, federated learning approaches can enable privacy-preserving model training, while real-time intrusion detection frameworks can enhance response speed in critical healthcare applications.

Moreover, future work should focus on the development of lightweight deep learning models to facilitate their usage in resource-constrained IoMT devices.

Further, the use of federated learning techniques can facilitate the development of intrusion detection systems without compromising the privacy of patient data by not requiring the sharing of data.

The development of real-time intrusion detection systems will be another area of focus to facilitate the timely response to intrusion detection in time-sensitive healthcare environments.

V. Conclusion

The increased use of Internet of Medical Things (IoMT) technologies has greatly enhanced the quality of healthcare services through real-time monitoring, diagnostics, and patient management. However, the rapid digitalization in the healthcare industry has posed major cybersecurity risks to the healthcare system.

The conventional intrusion detection systems have shown limited capabilities in dealing with the advanced cyber threats. Although the machine learning-based intrusion detection systems demonstrate promising results in solving the problem, they have some major drawbacks in terms of adaptability.

The deep learning-based intrusion detection systems have shown significant potential in solving the problem of cybersecurity in the healthcare system. Deep Learning algorithms such as Deep Neural Network, Convolution Neural Network, and Recurrent Neural Network have shown promising results in the detection of various types of cyber-attacks such as malware, botnet, and denial-of-service attacks [5].

Moreover, the integration of threat mitigation techniques with deep learning can enable the healthcare systems to address cyber threats in a dynamic manner. Such systems can provide better resilience against cyber threats.

However, there are challenges in the deployment of deep learning-based systems in the healthcare sector in terms of the complexity of the computations involved, privacy considerations, and the need for real-time responses. Additionally, there is a need to develop specific systems for the healthcare sector that can cater to the environment of IoMT devices.

Overall, the use of deep learning in the development of intrusion detection systems can provide a viable solution for the security of the modern healthcare sector. There is a need to develop efficient systems that can provide the necessary security to the healthcare sector in terms of the safety of the sensitive information of the patients.

VI. Acknowlegment

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Author Profile

Himara Nethmi Liyana Patabendi

BSc in Cyber Security (Undergraduate), Sri Lanka Institute of Information Technology (SLIIT)
it22580862@my.sliit.lk

Himara Nethmi Liyana Patabendi is currently pursuing a bachelor's degree in cyber security at the Sri Lanka Institute of Information Technology (SLIIT). With a deep interest in information security, penetration testing, network security, web security and security policy frameworks. She has gained practical experience in areas such as ethical hacking, network security, and incident response. She is passionate about exploring emerging trends in cybersecurity, particularly the vulnerabilities and protection strategies for healthcare IoT and IoMT systems.